

LeanForControl

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Chapter 1

Linear systems

1.1 Observability and controllability matrices

For a finite-dimensional linear system

$$\dot{x} = Ax + Bu, \quad y = Cx,$$

with state matrix A , input matrix B , and output matrix C , this section collects the (finite-horizon) controllability and observability matrices and the bridge theorem connecting the textbook observability predicate to the kernel of the observability matrix.

1.1.1 Observability matrix

Definition 1. The *observability matrix* of a pair (A, C) with $A \in \mathbb{F}^{n \times n}$ and $C \in \mathbb{F}^{p \times n}$ is the block-row matrix

$$\mathcal{O}(A, C) = \begin{bmatrix} C \\ C A \\ C A^2 \\ \vdots \\ C A^{n-1} \end{bmatrix} \in \mathbb{F}^{(np) \times n}.$$

Rows are indexed by $\text{Fin } n \times \text{Fin } p$, so that A^k is available without casting $k : \text{Fin } n$ through Fin.val .

Definition 2. A linear system (A, C) is *observable* when the only state $x \in \mathbb{F}^n$ for which

$$C A^k x = 0 \quad \text{for every } k = 0, 1, \dots, n-1$$

is the zero state. This phrasing does not mention $\mathcal{O}(A, C)$, so the bridge theorem 8 has real content.

Theorem 3. *Block-row entry shape: at row (k, i) and column j , the observability matrix coincides with the (i, j) entry of $C A^k$:*

$$\mathcal{O}(A, C)_{(k,i),j} = (C A^k)_{i,j}.$$

Proof. Holds definitionally by the encoding chosen for definition 1. □

Theorem 4. For every state $x \in \mathbb{F}^n$ and every $(k, i) \in \text{Fin } n \times \text{Fin } p$,

$$(\mathcal{O}(A, C) \cdot x)_{(k, i)} = (C A^k \cdot x)_i.$$

This is the definitional bridge that powers theorem 8.

Proof. Holds by definitional unfolding of $\mathcal{O}(A, C)$ (see definition 1). $\square$

1.1.2 Controllability matrix

Definition 5. The *controllability matrix* of a pair (A, B) with $A \in \mathbb{F}^{n \times n}$ and $B \in \mathbb{F}^{n \times m}$ is the block-column matrix

$$\mathcal{C}(A, B) = [B \quad AB \quad A^2 B \quad \dots \quad A^{n-1} B] \in \mathbb{F}^{n \times (nm)}.$$

Columns are indexed by $\text{Fin } n \times \text{Fin } m$, so that A^k is available without casting $k : \text{Fin } n$ through Fin.val .

Theorem 6. *Block-column entry shape:* at row i and column (k, j) , the controllability matrix coincides with the (i, j) entry of $A^k B$:

$$\mathcal{C}(A, B)_{i, (k, j)} = (A^k B)_{i, j}.$$

Proof. Holds definitionally by the encoding chosen for definition 5. $\square$

Definition 7. A linear system (A, B) is *controllable* when every target state $x \in \mathbb{F}^n$ is reachable from the origin in n steps: there exist input vectors $u_0, u_1, \dots, u_{n-1} \in \mathbb{F}^m$ such that

$$x = \sum_{k=0}^{n-1} A^k B u_k.$$

This phrasing does not name the controllability matrix, so the bridge theorem 10 has real content.

1.1.3 Milestone: observability iff trivial kernel of $\mathcal{O}$

Theorem 8. A finite-dimensional linear system (A, C) is observable in the sense of definition 2 if and only if the observability matrix $\mathcal{O}(A, C)$ has trivial kernel under matrix-vector multiplication:

$$\text{IsObservable}(A, C) \iff (\forall x, \mathcal{O}(A, C) \cdot x = 0 \Rightarrow x = 0).$$

Proof. Both directions follow from the bridge theorem 4, which identifies the (k, i) -coordinate of $\mathcal{O}(A, C) \cdot x$ with the i -coordinate of $(C A^k) \cdot x$. The forward direction reads the per-power kernel out of the assembled kernel coordinatewise; the reverse direction packs them back. $\square$

1.1.4 Rank characterizations

Theorem 9. A finite-dimensional system (A, C) is observable if and only if the observability matrix $\mathcal{O}(A, C)$ has full column rank, i.e.

$$\text{rank } \mathcal{O}(A, C) = n,$$

where n is the state dimension.

Proof. Chain the kernel-form milestone theorem 8 with the matrix-level bridge $(\forall x, M \cdot x = 0 \Rightarrow x = 0) \iff \text{rank } M = n$ from ‘MatrixAlgebra.Rank’, applied with $M = \mathcal{O}(A, C)$. $\square$

Theorem 10. *A linear system (A, B) is controllable if and only if the controllability matrix has full row rank,*

$$\text{rank } \mathcal{C}(A, B) = n,$$

where n is the state dimension.

Proof. Combine the reindexing identity $\mathcal{C}(A, B) \cdot u = \sum_k A^k B \cdot u_k$ (with $u_k(j) = u(k, j)$) with the matrix-level bridge $(\forall y, \exists x, Mx = y) \iff \text{rank } M = n$ from ‘MatrixAlgebra.Rank’, applied with $M = \mathcal{C}(A, B)$. $\square$

1.1.5 Hautus observability

Definition 11. The *unobservable subspace* of (A, C) is the A -invariant subspace

$$\mathcal{N}(A, C) = \{v \in \mathbb{C}^n : C A^k v = 0 \text{ for every } k = 0, 1, \dots, n-1\}.$$

Equivalently, $\mathcal{N}(A, C) = \ker \mathcal{O}(A, C)$, but here it is phrased without naming $\mathcal{O}$ so the bridge theorem 12 reads as content.

Theorem 12. *A finite-dimensional system (A, C) is observable in the sense of definition 2 if and only if its unobservable subspace is trivial:*

$$\mathcal{N}(A, C) = \{0\} \iff \text{IsObservable}(A, C).$$

Proof. Both directions are membership unfoldings of definition 11 against definition 2. $\square$

Theorem 13. *The unobservable subspace is closed under the action of A : for every $v \in \mathcal{N}(A, C)$, also $Av \in \mathcal{N}(A, C)$.*

Proof. For $k = 0, \dots, n-2$ this is direct from the definition. For $k = n-1$ we land at $CA^{n-1}v$, which Cayley-Hamilton rewrites as a $\mathbb{C}$ -linear combination of $CA^i v$ for $i < n$. Each of those is zero, so the combination is zero. $\square$

Definition 14. The *Hautus observability matrix* of (A, C) at a complex number μ is the block-row matrix

$$H_{A,C}(\mu) = \begin{bmatrix} \mu I - A \\ C \end{bmatrix} \in \mathbb{C}^{(n+p) \times n}.$$

Theorem 15. *For every $\mu \in \mathbb{C}$ and $v \in \mathbb{C}^n$,*

$$H_{A,C}(\mu) \cdot v = 0 \iff (Av = \mu v \wedge Cv = 0).$$

Proof. $\square$

Theorem 16. *If (A, C) is not observable, then there exists $\mu \in \mathbb{C}$ and a nonzero vector $v \in \mathbb{C}^n$ such that*

$$H_{A,C}(\mu) \cdot v = 0,$$

i.e. μ is an eigenvalue of A with an eigenvector in $\ker C$.

Proof. The unobservable subspace definition 11 is nontrivial by theorem 12. It is A -invariant by theorem 13, hence finite-dimensional and invariant; over $\mathbb{C}$ this guarantees a nonzero eigenvector inside it. That vector is automatically in $\ker C$ (membership at $k = 0$), so it lies in the kernel of $H_{A,C}(\mu)$ via theorem 15. $\square$

Theorem 17. *If $\mu \in \mathbb{C}$ and $v \neq 0$ satisfy $H_{A,C}(\mu) \cdot v = 0$, then (A, C) is not observable.*

Proof. The witness gives $Av = \mu v$ and $Cv = 0$. By induction on k , $A^k v = \mu^k v$, so $C A^k v = \mu^k Cv = 0$ for every k . The vector v is therefore in the unobservable subspace and is nonzero, so observability fails. $\square$

Theorem 18. *A finite-dimensional system (A, C) over $\mathbb{C}$ is observable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H_{A,C}(\mu)$ has trivial kernel:*

$$\text{IsObservable}(A, C) \iff \forall \mu \in \mathbb{C}, \ker H_{A,C}(\mu) = \{0\}.$$

Proof. Combine theorem 16 (failure direction) and theorem 17 (converse). $\square$

1.1.6 Hautus controllability

Definition 19. The *Hautus controllability matrix* of (A, B) at a complex number μ is the block-column matrix

$$H_{A,B}^{\text{ctrl}}(\mu) = [\mu I - A \quad B] \in \mathbb{C}^{n \times (n+m)}.$$

Theorem 20. *A finite-dimensional system (A, B) over $\mathbb{C}$ is controllable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H_{A,B}^{\text{ctrl}}(\mu)$ has full row rank:*

$$\text{IsControllable}(A, B) \iff \forall \mu \in \mathbb{C}, \text{rank } H_{A,B}^{\text{ctrl}}(\mu) = n.$$

Proof. Combine the duality bridge `isControllable_iff_isObservable_transpose` with the observability Hautus iff theorem 18, then convert the kernel-form RHS to the rank form via the matrix bridge `mulVec_kernel_trivial_iff_rank_eq_card_cols` from ‘MatrixAlgebra.Rank’, and finally identify the transposed Hautus matrices. $\square$

1.2 Reachable subspace and Kalman decomposition

Definition 21. The reachable subspace of a pair (A, B) is the column span of its controllability matrix:

$$\mathcal{R}(A, B) = \text{range } \mathcal{C}(A, B).$$

Thus it is exactly the span of $B, AB, \dots, A^{n-1}B$.

Theorem 22. *Let $A \in \mathbb{F}^{n \times n}$ and $B \in \mathbb{F}^{n \times m}$, and let $\mathcal{C}(A, B) \in \mathbb{F}^{n \times (nm)}$ be their controllability matrix. For $x \in \mathbb{F}^n$,*

$$x \in \mathcal{R}(A, B) \iff \exists u : \text{Fin}(n) \times \text{Fin}(m) \rightarrow \mathbb{F}, \quad \mathcal{C}(A, B)u = x.$$

Here $\mathcal{R}(A, B) = \text{range } \mathcal{C}(A, B)$ is the reachable subspace.

Proof. $\square$

Theorem 23. *A pair (A, B) is controllable if and only if its reachable subspace is the whole state space.*

Proof. □

Theorem 24. *The reachable subspace is A -invariant: $A\mathcal{R}(A, B) \subseteq \mathcal{R}(A, B)$.*

Proof. □

Definition 25. A Kalman decomposition of (A, B, C) records four coordinate sectors adapted to the reachable and unobservable subspaces. Their direct sums recover those subspaces and the full state space. The chosen complements are noncanonical and are not individually asserted to be invariant under A .

Theorem 26. *Every finite-dimensional complex state-space system has four coordinate sectors adapted to its reachable and unobservable subspaces. The complements are noncanonical vector-space complements and are not individually asserted to be invariant under A .*

Proof. □

Definition 27. Addition of the four Kalman coordinate sectors defines a linear equivalence $X_{c\bar{o}} \times X_{co} \times X_{\bar{c}\bar{o}} \times X_{\bar{c}o} \simeq \mathbb{C}^n$. Choosing sector bases therefore gives an adapted basis of the original state space.

Theorem 28. *In a basis adapted to the four Kalman coordinate sectors, ordered as controllable-unobservable, controllable-observable, uncontrollable-unobservable, uncontrollable-observable, the system has the forced-zero pattern*

$$A' = \begin{bmatrix} * & * & * & * \\ 0 & * & 0 & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{bmatrix}, \quad B' = \begin{bmatrix} * \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad C' = [0 \quad * \quad 0 \quad *].$$

Every displayed zero is asserted entrywise; no vanishing claim is made for the starred blocks. The chosen complement sectors are not individually asserted to be invariant under A .

Proof. Reachable-subspace invariance forces the lower-left state and input zeros, unobservable-subspace invariance forces the remaining state zeros, and the zeroth observability condition forces the output zeros. Coordinate-basis representation lemmas translate these component statements to individual matrix entries. □

1.3 Hurwitz stability

For a real square matrix A , Hurwitz stability is expressed through the complex eigenpairs of its entrywise complexification. The rate-indexed predicate records a strict spectral margin, and ordinary Hurwitz stability is the zero-rate case.

Definition 29. A real square matrix A is *Hurwitz with decay rate α* when every complex eigenpair (μ, v) with $v \neq 0$ satisfies $\text{Re}(\mu) < -\alpha$.

Definition 30. A real square matrix is *Hurwitz* when every complex eigenvalue has strictly negative real part.

Theorem 31. *A real matrix A is Hurwitz with decay rate α if and only if the spectrally shifted matrix $A + \alpha I$ is Hurwitz.*

Proof. A complex eigenvalue μ of A becomes $\mu + \alpha$ after the shift, and $\operatorname{Re}(\mu) < -\alpha$ is equivalent to $\operatorname{Re}(\mu + \alpha) < 0$. $\square$

1.4 Matrix Lyapunov equations and quadratic certificates

1.5 Peano–Baker series and the state transition matrix

For a genuinely time-varying state matrix $A(t)$, this section constructs the state transition matrix $\Phi(t, t_0)$ as the Peano–Baker series and proves Theorem 5.1: Φ solves the matrix ODE $\dot{\Phi}(t, t_0) = A(t)\Phi(t, t_0)$ with $\Phi(t_0, t_0) = I$, and consequently $x(t) := \Phi(t, t_0)x_0$ solves $\dot{x} = A(t)x$, $x(t_0) = x_0$.

Chapter 2

Ordinary differential equations

2.1 Dini derivatives

Definition 32. The *upper right Dini derivative* of $f : \mathbb{R} \rightarrow \mathbb{R}$ at t is

$$D^+f(t) = \limsup_{h \rightarrow 0^+} \frac{f(t+h) - f(t)}{h}.$$

When f is differentiable at t this equals $f'(t)$.

2.2 Gronwall–Bellman inequality

Theorem 33. Gronwall–Bellman inequality. Let $\Lambda, \mu : [a, b] \rightarrow \mathbb{R}$ be continuous with $\mu \geq 0$, and let $y : [a, b] \rightarrow \mathbb{R}$ be continuous satisfying

$$y(t) \leq \Lambda(t) + \int_a^t \mu(s) y(s) \, ds \quad \forall t \in [a, b].$$

Then

$$y(t) \leq \Lambda(t) + \int_a^t \Lambda(s) \mu(s) e^{\int_s^t \mu(\tau) \, d\tau} \, ds \quad \forall t \in [a, b].$$

Proof.

□

2.3 Continuous dependence on parameters

Definition 34. A function $x : [t_0, t_1] \rightarrow E$ is an *integral solution* of $\dot{x} = F(t, x)$ with initial value x_0 when

$$x(t) = x_0 + \int_{t_0}^t F(s, x(s)) \, ds \quad \forall t \in [t_0, t_1].$$

Theorem 35. *Theorem 3.5 (Continuous dependence on parameters). If y solves $\dot{y} = f(t, y)$ and z solves $\dot{z} = f(t, z) + g(t, z)$ with $\|g(t, x)\| \leq \alpha$ and $\|z_0 - y_0\| \leq \alpha$, and $\alpha(1 + 1/L)e^{L(t_1 - t_0)} \leq \varepsilon$, then $\|y(t) - z(t)\| \leq \varepsilon$ for all $t \in [t_0, t_1]$.*

Proof. Reduce to theorem 33 applied to $\|y - z\|$, using the L -Lipschitz bound on f and the α -bound on g . $\square$

2.4 Comparison lemma

Theorem 36. *Comparison Lemma 3.4. If u solves $\dot{u} = f(t, u)$ with $u(t_0) = u_0$, and v is continuous with $D^+v(t) \leq f(t, v(t))$ and $v(t_0) \leq u_0$, then $v(t) \leq u(t)$ for all $t \in [t_0, t_1]$.*

Proof. For each $\lambda > 0$, the perturbed solution z_λ of $\dot{z} = f(t, z) + \lambda$ satisfies $v \leq z_\lambda$ by ???. By theorem 35, $\|u - z_\lambda\| \leq \varepsilon/2$. Since $\varepsilon > 0$ is arbitrary, $v(t) \leq u(t)$. $\square$

Chapter 3

Lyapunov stability

3.1 Trajectories, equilibria, and comparison functions

Definition 37. A point $x_{\text{eq}} \in \mathbb{R}^n$ is an *equilibrium* of $\dot{x} = f(x)$ when $f(x_{\text{eq}}) = 0$.

Definition 38. A *class $\mathcal{K}$* function on $[0, a)$ is a continuous strictly increasing map $\alpha : [0, a) \rightarrow [0, b)$ with $\alpha(0) = 0$. The structure records both α and its inverse $\alpha^{-1} : [0, b) \rightarrow [0, a)$.

Definition 39. A *class $\mathcal{K}_\infty$* function is a continuous strictly increasing map $\alpha : [0, \infty) \rightarrow [0, \infty)$ with $\alpha(0) = 0$ and $\alpha(r) \rightarrow \infty$ as $r \rightarrow \infty$.

3.2 Stability predicates

The predicates below quantify over every solution segment on a compact interval $[t_0, t_1]$. This prevents finite escape from making a stability claim vacuous merely because a non-equilibrium solution is not globally defined. A solution segment is an integral curve of the vector field restricted to that interval, in the sense of `IsTrajectoryOn`.

Definition 40. The equilibrium x_{eq} is *Lyapunov stable* when

$$\forall \varepsilon > 0, \exists \delta > 0, \forall \varphi, \text{IsTrajectory}(\varphi, f) \Rightarrow \|\varphi(0) - x_{\text{eq}}\| < \delta \Rightarrow \forall t \geq 0, \|\varphi(t) - x_{\text{eq}}\| < \varepsilon.$$

Definition 41. The equilibrium x_{eq} is *locally asymptotically stable* (LAS) when it is Lyapunov stable (definition 40) and there exists $c > 0$ such that every trajectory φ with $\|\varphi(0) - x_{\text{eq}}\| < c$ satisfies $\varphi(t) \rightarrow x_{\text{eq}}$ as $t \rightarrow \infty$.

Definition 42. The equilibrium x_{eq} is *globally asymptotically stable* (GAS) when it is Lyapunov stable (definition 40) and every trajectory φ satisfies $\varphi(t) \rightarrow x_{\text{eq}}$ as $t \rightarrow \infty$.

3.3 Sublevel sets and positive invariance

Definition 43. The *sublevel set* of $V : \mathbb{R}^n \rightarrow \mathbb{R}$ at level $c \in \mathbb{R}$ is

$$\Omega_c(V) = \{x \in \mathbb{R}^n : V(x) \leq c\}.$$

Definition 44. A set $S \subseteq \mathbb{R}^n$ is *positively invariant* for $\dot{x} = f(x)$ when every trajectory φ starting in S remains in S for all future time:

$$\varphi(0) \in S \Rightarrow \varphi(t) \in S \quad \forall t \geq 0.$$

Theorem 45. If $V : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous and radially unbounded ($V(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$), then every sublevel set $\Omega_c(V)$ (definition 43) is compact.

Proof. Closed: $\Omega_c(V) = V^{-1}((-\infty, c])$ by continuity. Bounded: coercivity yields R with $\Omega_c(V) \subseteq \overline{B}(0, R)$. Compact: Heine–Borel in $\mathbb{R}^n$. $\square$

3.4 Lyapunov functions

Definition 46. A function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ is a *local Lyapunov function* on an open domain $D \ni x_{\text{eq}}$ when V is smooth, $V(x_{\text{eq}}) = 0$, $V > 0$ on $D \setminus \{x_{\text{eq}}\}$, and the Lie derivative satisfies $\dot{V}(x) = \nabla V(x) \cdot f(x) \leq 0$ for all $x \in D$.

Definition 47. A *strict local Lyapunov function* on D strengthens definition 46: the Lie derivative satisfies $\dot{V}(x) < 0$ for all $x \in D \setminus \{x_{\text{eq}}\}$, and there exists $c > 0$ such that the compact sublevel set $\Omega_c(V) \subseteq D$ (definition 43).

Definition 48. A *global strict Lyapunov function* for $\dot{x} = f(x)$ at x_{eq} is a C^1 map $V : \mathbb{R}^n \rightarrow \mathbb{R}$ with $V(x_{\text{eq}}) = 0$, $V > 0$ everywhere else, $\dot{V}(x) < 0$ on $\mathbb{R}^n \setminus \{x_{\text{eq}}\}$, and all sublevel sets $\Omega_c(V)$ compact (coercivity).

Definition 49. The classical GAS Lyapunov certificate: a C^1 map $V : \mathbb{R}^n \rightarrow \mathbb{R}$ with $V(x_{\text{eq}}) = 0$, $V > 0$ elsewhere, $\dot{V} < 0$ on $\mathbb{R}^n \setminus \{x_{\text{eq}}\}$, and radially unbounded ($V(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$). Implies definition 48 via theorem 45.

3.5 Milestone theorems for autonomous systems

Theorem 50. *Lyapunov’s stability theorem.* If V is a local Lyapunov function (definition 46) for $\dot{x} = f(x)$ on a domain $D \ni x_{\text{eq}}$, then x_{eq} is Lyapunov stable (definition 40).

Proof. Pick ε_0 so $\overline{B}(x_{\text{eq}}, \varepsilon_0) \subseteq D$. Let $m = \min_{S_{\varepsilon_0}} V > 0$. Choose δ with $V < m$ on $B(x_{\text{eq}}, \delta)$. If $\|\varphi(t^*) - x_{\text{eq}}\| \geq \varepsilon$, monotonicity of V gives $V(\varphi(t^*)) \leq V(\varphi(0)) < m \leq V(\varphi(t^*))$, a contradiction. $\square$

Theorem 51. *Lyapunov’s local asymptotic stability theorem.* If V is a strict local Lyapunov function (definition 47) and f is continuous, then x_{eq} is locally asymptotically stable (definition 41).

Proof. The compact sublevel set $\Omega_{c_0} \subseteq D$ is positively invariant. $V(\varphi(t)) \rightarrow L \geq 0$ by monotone convergence; $L = 0$ by compactness; $\varphi(t) \rightarrow x_{\text{eq}}$. $\square$

Theorem 52. *Lyapunov’s global asymptotic stability theorem.* If V is a global strict Lyapunov function (definition 48) and f is continuous, then x_{eq} is globally asymptotically stable (definition 42).

Proof. Stability from theorem 50. For convergence: $V(\varphi(t)) \rightarrow L \geq 0$ by monotone convergence; $L = 0$ by a LaSalle-type argument; $\varphi(t) \rightarrow x_{\text{eq}}$ by coercivity. $\square$

Theorem 53. Corollary. *If V is a radially unbounded strict Lyapunov function (definition 49) and f is continuous, then x_{eq} is globally asymptotically stable.*

Proof. Radial unboundedness gives compact sublevel sets (theorem 45), so V satisfies definition 48; apply theorem 52. $\square$

3.6 Lyapunov's indirect method

The stable-branch theorem gives uniform local exponential decay on every finite forward solution segment, without silently excluding solutions that are not globally defined.

3.7 LaSalle's invariance principle

Theorem 54. LaSalle's invariance principle. *Let Ω be compact and positively invariant for $\dot{x} = f(x)$, $V \in C^1$ with $\dot{V}(x) \leq 0$ on Ω , and $M \subseteq \Omega$ closed. If the ω -limit set of any trajectory φ starting in Ω satisfies $\omega(\varphi) \subseteq M$, then $\varphi(t) \rightarrow M$ as $t \rightarrow \infty$.*

Proof. $\square$

Theorem 55. Barbashin's theorem. *If $V \in C^1$ is positive definite on D , $\dot{V} \leq 0$ on D , there exists a compact sublevel set $\Omega_c \subseteq D$, and every trajectory starting in Ω_c has ω -limit set $\subseteq \{x_{\text{eq}}\}$, then x_{eq} is locally asymptotically stable (definition 41).*

Proof. $\square$

Theorem 56. Krasovskii's theorem. *If $V \in C^1$ is positive definite, $\dot{V} \leq 0$ on $\mathbb{R}^n$, radially unbounded, and every trajectory has ω -limit set $\subseteq \{x_{\text{eq}}\}$, then x_{eq} is globally asymptotically stable (definition 42).*

Proof. $\square$

3.8 Lyapunov class $\mathcal{K}$ bounds

Theorem 57. Class $\mathcal{K}$ sandwich bounds. *For any continuous positive-definite function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ on $\overline{B}(0, r)$, there exist class $\mathcal{K}$ functions α_1, α_2 such that*

$$\alpha_1(\|x\|) \leq V(x) \leq \alpha_2(\|x\|) \quad \forall \|x\| \leq r.$$

Proof. Construct α_1 from the infimum of V on annuli (lower bound), and α_2 from the supremum of V on balls (upper bound). Each is sandwiched by a class $\mathcal{K}$ function via the axioms in `LeanForControl.axioms`. $\square$

3.9 Class $\mathcal{KL}$ functions and the Osgood construction

Definition 58. A class $\mathcal{KL}$ function on $[0, a) \times [0, \infty)$ is continuous, class $\mathcal{K}$ in the first argument, and for each fixed $r > 0$ is strictly decreasing and tends to 0 as $s \rightarrow \infty$. It arises as the bound $\|x(t)\| \leq \beta(\|x_0\|, t)$ in asymptotic stability estimates.

Theorem 59. Osgood construction. Given a class $\mathcal{K}$ function α satisfying the Osgood condition $\int_0^\varepsilon \frac{1}{\alpha(x)} dx = +\infty$, the function $\sigma(r, s) = \eta^{-1}(\eta(r) + s)$, where $\eta(y) = -\int_{\text{base}}^y \frac{1}{\alpha(x)} dx$, is a class $\mathcal{KL}$ function (definition 58).

Proof.

□